

Computer Organization & Architecture (COA)

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 122

Question 1 of 122 | GATE 2025 Set 1 (CS1) | Q.11 (Computer Science & IT)

Q.11

Suppose a program is running on a non-pipelined single processor computer system.

The computer is connected to an external device that can interrupt the processor asynchronously. The processor needs to execute the interrupt service routine (ISR) to serve this interrupt. The following steps (not necessarily in order) are taken by the processor when the interrupt arrives:

(i)

The processor saves the content of the program counter.

(ii)

The program counter is loaded with the start address of the ISR.

(iii) The processor finishes the present instruction.

Which ONE of the following is the CORRECT sequence of steps?

(A) (iii), (i), (ii)

(B) (i), (iii), (ii)

(C) (i), (ii), (iii)

(D) (iii), (ii), (i)

Question 2 of 122 | GATE 2025 Set 1 (CS1) | Q.21 (Computer Science & IT)

Q.21 Consider the following B+ tree with 5 nodes, in which a node can store at most 3 key values.

The value 23 is now inserted in the B+ tree. Which of the following options(s) is/are CORRECT?

(A) None of the nodes will split.

(B) At least one node will split and redistribute.

(C) The total number of nodes will remain same.

(D) The height of the tree will increase.

Question 3 of 122 | GATE 2025 Set 1 (CS1) | Q.27 (Computer Science & IT)

Q.27 A partial data path of a processor is given in the figure, where RA, RB, and RZ are 32-bit registers. Which option(s) is/are CORRECT related to arithmetic operations using the data path as shown?

(A) The data path can implement arithmetic operations involving two registers.

- (B) The data path can implement arithmetic operations involving one register and one immediate value.
- (C) The data path can implement arithmetic operations involving two immediate values.
- (D) The data path can only implement arithmetic operations involving one register and one immediate value.

Question 4 of 122 | GATE 2025 Set 1 (CS1) | Q.36 (Computer Science & IT)

Q.36

Consider a memory system with 1M bytes of main memory and 16K bytes of cache memory. Assume that the processor generates 20-bit memory address, and the cache block size is 16 bytes. If the cache uses direct mapping, how many bits will be required to store all the tag values? [Assume memory is byte addressable, $1K=2^{10}$, $1M=2^{20}$.]

- (A) 6×2^{10}
- (B) 8×2^{10}
- (C) 212
- (D) 214

Question 5 of 122 | GATE 2025 Set 1 (CS1) | Q.42 (Computer Science & IT)

Q.42

Consider the following four variable Boolean function in sum-of-product form

$$F(b_3, b_2, b_1, b_0) = \sum(0, 2, 4, 8, 10, 11, 12).$$

where the value of the function is computed by considering $b_3b_2b_1b_0$ as a 4-bit binary number, where b_3 denotes the most significant bit and b_0 denotes the least significant bit. Note that there are no don't care terms. Which ONE of the following options is the CORRECT minimized Boolean expression for F ?

- (A) $b_1b_0 + b_2b_0 + b_1b_2b_3$
- (B) $b_1b_0 + b_2b_0$
- (C) $b_2b_0 + b_1b_2b_3$
- (D) $b_0b_2 + b_3$

Question 6 of 122 | GATE 2025 Set 1 (CS1) | Q.43 (Computer Science & IT)

Q.43

Let $G(V, E)$ be an undirected and unweighted graph with 100 vertices. Let $d(u, v)$ denote the number of edges in a shortest path between vertices u and v in V . Let the maximum value of $d(u, v)$, $u, v \in V$ such that $u \neq v$, be 30. Let T be any breadth-first-search tree of G . Which ONE of the given options is CORRECT for every such graph G ?

- (A) The height of T is exactly 15.
- (B) The height of T is exactly 30.
- (C) The height of T is at least 15.
- (D) The height of T is at least 30.

Question 7 of 122 | GATE 2025 Set 1 (CS1) | Q.55 (Computer Science & IT)

Q.55

Consider the following database tables of a sports league.

player(pid,pname,age)

coach(cid,cname)

team(tid,tname,city,cid)

members(pid,tid)

An instance of the table and an SQL query are given.

player coach team members

```
SELECT MIN(P.age)
```

```
FROM player P
```

```
WHERE P.pid IN (
```

```
SELECT M.pid
```

```
FROM team T, coach C, members M
```

```
WHERE C.cname = 'Mark'
```

```
AND T.cid = C.cid
```

```
AND M.tid = T.tid
```

```
)
```

The value returned by the given SQL query is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 8 of 122 | GATE 2025 Set 1 (CS1) | Q.60 (Computer Science & IT)

Q.60

Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning back to the initial state is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 9 of 122 | GATE 2025 Set 1 (CS1) | Q.61 (Computer Science & IT)

Q.61

```
#include <stdio.h>
int foo(int S[],int size){
if(size == 0) return 0;
```

```

if(size == 1) return 1;
if(S[0] != S[1]) return 1+foo(S+1,size-1);
return foo(S+1,size-1);
}
int main(){
int A[]={0,1,2,2,2,0,0,1,1};
printf("%d",foo(A,9));
return 0;
}

```

The value printed by the given C program is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 10 of 122 | GATE 2025 Set 1 (CS1) | Q.63 (Computer Science & IT)

Q.63

Consider the following C program:

```

#include <stdio.h>
int gate (int n) {
int d, t, newnum, turn;
newnum = turn = 0; t=1;
while (n>=t) t *= 10;
t /=10;
while (t>0) {
d = n/t;
n = n%t;
t /= 10;
if (turn) newnum = 10*newnum + d;
turn = (turn + 1) % 2;
}
return newnum;
}
int main () {
printf ("%d", gate(14362));
return 0;
}

```

The value printed by the given C program is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 11 of 122 | GATE 2025 Set 1 (CS1) | Q.64 (Computer Science & IT)

Q.64

The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is _____. (answer in

integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 12 of 122 | GATE 2025 Set 2 (CS2) | Q.27 (Computer Science & IT)

Q.27

An audit of a banking transactions system has found that on an earlier occasion, two joint holders of account A attempted simultaneous transfers of Rs. 10000 each from account A to account B. Both transactions read the same value, Rs. 11000, as the initial balance in A and were allowed to go through. B was credited Rs. 10000 twice. A was debited only once and ended up with a balance of Rs. 1000.

Which of the following properties is/are certain to have been violated by the system?

- (A) Atomicity
- (B) Consistency
- (C) Isolation
- (D) Durability

Question 13 of 122 | GATE 2025 Set 2 (CS2) | Q.28 (Computer Science & IT)

Q.28

Which of the following is/are part of an Instruction Set Architecture of a processor?

- (A) The size of the cache memory
- (B) The clock frequency of the processor
- (C) The number of cache memory levels
- (D) The total number of registers

Question 14 of 122 | GATE 2025 Set 2 (CS2) | Q.32 (Computer Science & IT)

Q.32

The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's algorithm:

M: 1100 1101 1110 1101 and Q: 1010 0100 1010 1010

The total number of addition and subtraction operations to be performed is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 15 of 122 | GATE 2025 Set 2 (CS2) | Q.35 (Computer Science & IT)

Q.35

Suppose the values 10, -4, 15, 30, 20, 5, 60, 19 are inserted in that order into an initially empty binary search tree. Let T be the resulting binary search tree. The number of edges in the path from the node containing 19 to the root node of T is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 16 of 122 | GATE 2025 Set 2 (CS2) | Q.39 (Computer Science & IT)

Q.39

For a direct-mapped cache, 4 bits are used for the tag field and 12 bits are used to index into a cache block. The size of each cache block is one byte. Assume that there is no other information stored for each cache block.

Which ONE of the following is the CORRECT option for the sizes of the main memory and the cache memory in this system (byte addressable), respectively?

- (A) 64 KB and 4 KB
- (B) 128 KB and 16 KB
- (C) 64 KB and 8 KB
- (D) 128 KB and 6 KB

Question 17 of 122 | GATE 2025 Set 2 (CS2) | Q.45 (Computer Science & IT)

Q.45

Consider a stack data structure into which we can PUSH and POP records. Assume that each record pushed in the stack has a positive integer key and that all keys are distinct.

We wish to augment the stack data structure with an $O(1)$ time MIN operation that returns a pointer to the record with smallest key present in the stack

- 1) without deleting the corresponding record, and
- 2) without increasing the complexities of the standard stack operations.

Which one or more of the following approach(es) can achieve it?

- (A) Keep with every record in the stack, a pointer to the record with the smallest key below it.
- (B) Keep a pointer to the record with the smallest key in the stack.
- (C) Keep an auxiliary array in which the key values of the records in the stack are maintained in sorted order.
- (D) Keep a Min-Heap in which the key values of the records in the stack are maintained.

Question 18 of 122 | GATE 2025 Set 2 (CS2) | Q.49 (Computer Science & IT)

Q.49

Three floating point numbers X, Y, and Z are stored in three registers RX, RY, and RZ, respectively in IEEE 754 single precision format as given below in

hexadecimal:

$RX = 0xC1100000$, $RY = 0x40C00000$, and $RZ = 0x41400000$

Which of the following option(s) is/are CORRECT?

- (A) $4(X + Y) + Z = 0$
- (B) $2Y - Z = 0$
- (C) $4X + 3Z = 0$
- (D) $X + Y + Z = 0$

Question 19 of 122 | GATE 2025 Set 2 (CS2) | Q.55 (Computer Science & IT)

Q.55

Given a computing system with two levels of cache (L1 and L2) and a main memory. The first level (L1) cache access time is 1 nanosecond (ns) and the "hit rate" for L1 cache is 90% while the processor is accessing the data from L1 cache. Whereas, for the second level (L2) cache, the "hit rate" is 80% and the "miss penalty" for transferring data from L2 cache to L1 cache is 10 ns. The "miss penalty" for the data to be transferred from main memory to L2 cache is 100 ns. Then the average memory access time in this system in nanoseconds is _____. (rounded off to one decimal place)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 20 of 122 | GATE 2025 Set 2 (CS2) | Q.56 (Computer Science & IT)

Q.56

A 5-stage instruction pipeline has stage delays of 180, 250, 150, 170, and 250, respectively, in nanoseconds. The delay of an inter-stage latch is 10 nanoseconds. Assume that there are no pipeline stalls due to branches and other hazards. The time taken to process 1000 instructions in microseconds is _____. (rounded off to two decimal places)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 21 of 122 | GATE 2025 Set 2 (CS2) | Q.57 (Computer Science & IT)

Q.57

In a B+- tree where each node can hold at most four key values, a root to leaf path consists of the following nodes:

$A = (49, 77, 83, -)$, $B = (7, 19, 33, 44)$, $C = (20^*, 22^*, 25^*, 26^*)$

The *-marked keys signify that these are data entries in a leaf.

Assume that a pointer between keys k_1 and k_2 points to a subtree containing keys in $[k_1, k_2)$, and that when a leaf is created, the smallest key in it is copied up into its parent.

A record with key value 23 is inserted into the B+- tree.

The smallest key value in the parent of the leaf that contains 25^* is _____.

(Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 22 of 122 | GATE 2025 Set 2 (CS2) | Q.61 (Computer Science & IT)

Q.61

An application executes 6.4×10^8 number of instructions in 6.3 seconds. There are four types of instructions, the details of which are given in the table. The duration of a clock cycle in nanoseconds is _____. (rounded off to one decimal place)

Instruction type

Clock cycles required per
instruction (CPI)

Number of instructions
executed

Branch

2

2.25×10^8

Load

5

1.20×10^8

Store

4

1.65×10^8

Arithmetic

3

1.30×10^8

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 23 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.14 (Computer Science & IT)

Q.14

Consider a permutation sampled uniformly at random from the set of all permutations of $\{1, 2, 3, \dots, n\}$ for some $n \geq 4$. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4. Which one of the following statements is TRUE?

- (A) The events X and Y are mutually exclusive
 - (B) The events X and Y are independent
 - (C) Either event X or Y must occur
 - (D) Event X is more likely than event Y
- Organizing Institute: IISc Bengaluru

Question 24 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.15 (Computer Science & IT)

Q.15

Which one of the following statements is FALSE?

- (A) In the cycle stealing mode of DMA, one word of data is transferred between an I/O device and main memory in a stolen cycle
 - (B) For bulk data transfer, the burst mode of DMA has a higher throughput than the cycle stealing mode
 - (C) Programmed I/O mechanism has a better CPU utilization than the interrupt driven I/O mechanism
 - (D) The CPU can start executing an interrupt service routine faster with vectored interrupts than with non-vectored interrupts
- Organizing Institute: IISc Bengaluru

Question 25 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.16 (Computer Science & IT)

Q.16

A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.

(i)

HTTP GET request for the index page

(ii)

DNS request to resolve the web server's name to its IP address

(iii)

HTTP GET request for an image object

(iv)

TCP SYN to open a connection to the web server

Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer ?

- (A) (iv), (ii), (iii), (i)
- (B) (ii), (iv), (iii), (i)
- (C) (ii), (iv), (i), (iii)
- (D) (iv), (ii), (i), (iii)

Question 26 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.17 (Computer Science & IT)

Q.17

Given an integer array of size N , we want to check if the array is sorted (in either ascending or descending order). An algorithm solves this problem by making a single pass through the array and comparing each element of the array only with its adjacent elements. The worst-case time complexity of this algorithm is

- (A) both $O(N)$ and $\Omega(N)$
- (B) $O(N)$ but not $\Omega(N)$
- (C) $\Omega(N)$ but not $O(N)$
- (D) neither $O(N)$ nor $\Omega(N)$ Organizing Institute: IISc Bengaluru

Question 27 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.18 (Computer Science & IT)

Q.18

Consider the following C program:

```
#include <stdio.h>
int main(){
    int a = 6;
    int b = 0;
    while(a < 10) {
        a = a / 12 + 1;
        a += b;}
    printf("%d", a);
    return 0;}
```

Which one of the following statements is CORRECT?

- (A) The program prints 9 as output
- (B) The program prints 10 as output
- (C) The program gets stuck in an infinite loop
- (D) The program prints 6 as output Organizing Institute: IISc Bengaluru

Question 28 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.19 (Computer Science & IT)

Q.19

Consider the following C program:

```
#include <stdio.h>
void fX();
int main(){
    fX();
    return 0;}
void fX(){
    char a;
    if((a=getchar()) != '\n')
        fX();
    if(a != '\n')
        putchar(a);}
```

Assume that the input to the program from the command line is 1234 followed by a newline character. Which one of the following statements is CORRECT?

- (A) The program will not terminate
 - (B) The program will terminate with no output
 - (C) The program will terminate with 4321 as output
 - (D) The program will terminate with 1234 as output
- Organizing Institute: IISc Bengaluru

Question 29 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.20 (Computer Science & IT)

Q.20

Let S be the specification: "Instructors teach courses. Students register for courses. Courses are allocated classrooms. Instructors guide students." Which one of the following ER diagrams CORRECTLY represents S?

- (A) (i)
 - (B) (ii)
 - (C) (iii)
 - (D) (iv)
- Organizing Institute: IISc Bengaluru

Question 30 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.22 (Computer Science & IT)

Q.22

Which of the following statements about a relation R in first normal form (1NF) is/are TRUE ?

- (A) R can have a multi-attribute key
 - (B) R cannot have a foreign key
 - (C) R cannot have a composite attribute
 - (D) R cannot have more than one candidate key
- Organizing Institute: IISc Bengaluru

Question 31 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.28 (Computer Science & IT)

Q.28

Consider the circuit shown below where the gates may have propagation delays. Assume that all signal transitions occur instantaneously and that wires have no delays. Which of the following statements about the circuit is/are CORRECT?

- (A) With no propagation delays, the output Y is always logic Zero
- (B) With no propagation delays, the output Y is always logic One
- (C) With propagation delays, the output Y can have a transient logic One after X transitions from logic Zero to logic One

- (D) With propagation delays, the output Y can have a transient logic Zero after X transitions from logic One to logic Zero
Organizing Institute: IISc Bengaluru

Question 32 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.30 (Computer Science & IT)

Q.30

Consider a 5-stage pipelined processor with Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Register Writeback (WB) stages. Which of the following statements about forwarding is/are CORRECT?

- (A) In a pipelined execution, forwarding means the result from a source stage of an earlier instruction is passed on to the destination stage of a later instruction
- (B) In forwarding, data from the output of the MEM stage can be passed on to the input of the EX stage of the next instruction
- (C) Forwarding cannot prevent all pipeline stalls
- (D) Forwarding does not require any extra hardware to retrieve the data from the pipeline stages
Organizing Institute: IISc Bengaluru

Question 33 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.32 (Computer Science & IT)

Q.32

Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from $A \times A$ to $A \sqcup B$. The number of possible values of $|A|$ is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 34 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.33 (Computer Science & IT)

Q.33

Consider the operator precedence and associativity rules for the integer arithmetic operators given in the table below.

Operator	
Precedence	
Associativity	
+	
Highest	
Left	
–	
High	
Right	
□	
Medium	
Right	

/

Low

Right

The value of the expression $3 + 1 + 5 \div 7 + 2 - 4 - 7 - 6 \div 2$ as per the above rules is _____

Organizing Institute: IISc Bengaluru

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 35 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.35 (Computer Science & IT)**

Q.35

Consider the following two relations, R(A,B) and S(A,C):

R

S

A

B

A

C

10

20

10

90

20

30

30

45

30

40

40

80

30

50

50

95

The total number of tuples obtained by evaluating the following expression

 $\sigma_{B < C}(R \bowtie R.A = S.AS)$

is _____

Organizing Institute: IISc Bengaluru

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 36 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.36 (Computer Science & IT)**

Q.36

Consider a network path $P \rightarrow Q \rightarrow R$ between nodes P and R via router Q . Node P sends a file of size 106 bytes to R via this path by splitting the file into chunks of 103 bytes each. Node P sends these chunks one after the other without any wait time between the successive chunk transmissions. Assume that the size of extra headers added to these chunks is negligible, and that the chunk size is less than the MTU. Each of the links $P \rightarrow Q$ and $Q \rightarrow R$ has a bandwidth of 106 bits/sec, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R , with negligible processing and queueing delays. Router Q can simultaneously receive on link $P \rightarrow Q$ and transmit on link $Q \rightarrow R$. Assume P starts transmitting the chunks at time $t = 0$. Which one of the following options gives the time (in seconds, rounded off to 3 decimal places) at which R receives all the chunks of the file?

- (A) 8.000
 - (B) 8.008
 - (C) 15.992
 - (D) 16.000
- Organizing Institute: IISc Bengaluru

Question 37 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.41 (Computer Science & IT)

Q.41

An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?

- (A) 82, 90, 101
 - (B) 82, 11, 93
 - (C) 131, 11, 93
 - (D) 131, 111, 90
- Organizing Institute: IISc Bengaluru

Question 38 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.43 (Computer Science & IT)

Q.43

Consider a binary min-heap containing 105 distinct elements. Let k be the index (in the underlying array) of the maximum element stored in the heap. The number of possible values of k is

- (A) 53
 - (B) 52
 - (C) 27
 - (D) 1
- Organizing Institute: IISc Bengaluru

Question 39 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.45 (Computer Science & IT)

Q.45

Let G be a directed graph and T a depth first search (DFS) spanning tree in G that is rooted at a vertex v . Suppose T is also a breadth first search (BFS) tree in G , rooted at v . Which of the following statements is/are TRUE for every such graph G and tree T ?

- (A) There are no back-edges in G with respect to the tree T
- (B) There are no cross-edges in G with respect to the tree T
- (C) There are no forward-edges in G with respect to the tree T
- (D) The only edges in G are the edges in T

Question 40 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.47 (Computer Science & IT)

Q.47

Consider a Boolean expression given by $F(X, Y, Z) = \sum(3, 5, 6, 7)$. Which of the following statements is/are CORRECT?

- (A) $F(X, Y, Z) = \prod(0, 1, 2, 4)$
- (B) $F(X, Y, Z) = XY + YZ + XZ$
- (C) $F(X, Y, Z)$ is independent of input Y
- (D) $F(X, Y, Z)$ is independent of input X

Question 41 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.48 (Computer Science & IT)

Q.48

Consider the following C function definition.

```
int f(int x, int y) {
    for (int i=0; i<y; i++) {
        x=x+x+y;
    }
    return x;
}
```

Which of the following statements is/are TRUE about the above function?

- (A) If the inputs are $x=20, y=10$, then the return value is greater than 220
- (B) If the inputs are $x=20, y=20$, then the return value is greater than 220
- (C) If the inputs are $x=20, y=10$, then the return value is less than 210
- (D) If the inputs are $x=10, y=20$, then the return value is greater than 220

Question 42 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.52 (Computer Science & IT)

Q.52

Consider the operators $\diamond$ and $\square$ defined by $a \diamond b = a + 2b$, $a \square b = ab$, for positive integers. Which of the following statements is/are TRUE?

- (A) Operator $\diamond$ obeys the associative law
- (B) Operator $\square$ obeys the associative law
- (C) Operator $\diamond$ over the operator $\square$ obeys the distributive law
- (D) Operator $\square$ over the operator $\diamond$ obeys the distributive law

Organizing Institute: IISc Bengaluru

Question 43 of 122 | GATE 2024 Set 1 (CS1 S5) | Q.56 (Computer Science & IT)

Q.56

A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stalls is 2. The program exhibits 2% miss rate on instruction cache and 8% miss rate on data cache. The miss penalty is 100 cycles. The speedup (rounded off to two decimal places) achieved with a perfect cache (i.e., with NO data or instruction cache misses) is _____

Organizing Institute: IISc Bengaluru

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 44 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.11 (Computer Science & IT)

Q.11

Consider a computer with a 4 MHz processor. Its DMA controller can transfer 8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following is the data transfer rate (in bits per second) of the DMA controller if 1% of the processor cycles are used for DMA?

- (A) 2,56,000
- (B) 3,200
- (C) 25,60,000
- (D) 32,000

Question 45 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.12 (Computer Science & IT)

Q.12

Let p and q be the following propositions:

p : Fail grade can be given.

q : Student scores more than 50% marks.

Consider the statement: "Fail grade cannot be given when student scores more than 50% marks."

Which one of the following is the CORRECT representation of the above statement

in propositional logic?

- (A) $q \rightarrow \neg \Box p$
- (B) $q \rightarrow p$
- (C) $p \rightarrow q$
- (D) $\neg \Box p \rightarrow q$

Question 46 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.13 (Computer Science & IT)

Q.13

Consider the following C program. Assume parameters to a function are evaluated from right to left.

```
#include <stdio.h>
int g(int p) { printf("%d", p); return p; }
int h(int q) { printf("%d", q); return q; }
void f(int x, int y) {
    g(x);
    h(y);
}
int main() {
    f(g(10),h(20));
}
```

Which one of the following options is the CORRECT output of the above C program?

- (A) 20101020
- (B) 10202010
- (C) 20102010
- (D) 10201020

Question 47 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.14 (Computer Science & IT)

Q.14

The format of a single-precision floating-point number as per the IEEE 754 standard is:

Sign
(1bit)
Exponent
(8 bits)
Mantissa
(23 bits)

Choose the largest floating-point number among the following options.

- (A) Sign Exponent Mantissa 0 0111 1111 1111 1111 1111 1111 111
 - (B) Sign Exponent Mantissa 0 1111 1110 1111 1111 1111 1111 111
 - (C) Sign Exponent Mantissa 0 1111 1111 1111 1111 1111 1111 111
 - (D) Sign Exponent Mantissa 0 0111 1111 0000 0000 0000 0000 000
- Organizing Institute: IISc, Bengaluru

Question 48 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.20 (Computer Science & IT)

Q.20

In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is TRUE?

- (A) The weak entity set MUST have total participation in the identifying relationship
 - (B) The owner entity set MUST have total participation in the identifying relationship
 - (C) Both weak and owner entity sets MUST have total participation in the identifying relationship
 - (D) Neither weak entity set nor owner entity set MUST have total participation in the identifying relationship
- Organizing Institute: IISc, Bengaluru

Question 49 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.28 (Computer Science & IT)

Q.28

Which of the following statements about IPv4 fragmentation is/are TRUE?

- (A) The fragmentation of an IP datagram is performed only at the source of the datagram
 - (B) The fragmentation of an IP datagram is performed at any IP router which finds that the size of the datagram to be transmitted exceeds the MTU
 - (C) The reassembly of fragments is performed only at the destination of the datagram
 - (D) The reassembly of fragments is performed at all intermediate routers along the path from the source to the destination
- Organizing Institute: IISc, Bengaluru

Question 50 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.30 (Computer Science & IT)

Q.30

For a Boolean variable x , which of the following statements is/are FALSE?

- (A) $x \cdot 1 = x$
 - (B) $x + 1 = x$
 - (C) $x \cdot x = 0$
 - (D) $x + \bar{x} = 1$
- Organizing Institute: IISc, Bengaluru

Question 51 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.31 (Computer Science & IT)

Q.31

An instruction format has the following structure:

Instruction Number: Opcode destination reg, source reg-1, source reg-2

Consider the following sequence of instructions to be executed in a pipelined processor:

I1: DIV R3, R1, R2

I2: SUB R5, R3, R4

I3: ADD R3, R5, R6

I4: MUL R7, R3, R8

Which of the following statements is/are TRUE?

- (A) There is a RAW dependency on R3 between I1 and I2
 - (B) There is a WAR dependency on R3 between I1 and I3
 - (C) There is a RAW dependency on R3 between I2 and I3
 - (D) There is a WAW dependency on R3 between I3 and I4
- Organizing Institute: IISc, Bengaluru

Question 52 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.33 (Computer Science & IT)

Q.33

Consider the following C function definition.

```
int fX(char *a){
char *b = a;
while(*b)
b++;
return b - a;}

```

Which of the following statements is/are TRUE?

- (A) The function call fX("abcd") will always return a value
 - (B) Assuming a character array c is declared as char c[] = "abcd" in main(), the function call fX(c) will always return a value
 - (C) The code of the function will not compile
 - (D) Assuming a character pointer c is declared as char *c = "abcd" in main(), the function call fX(c) will always return a value
- Organizing Institute: IISc, Bengaluru

Question 53 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.35 (Computer Science & IT)

Q.35

Let $A[]$ be an array containing integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in non-decreasing order. The distance of the array [2, 5, 3, 1, 4, 2, 6] is _____

Organizing Institute: IISc, Bengaluru

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 54 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.36 (Computer Science & IT)

Q.36

What is the output of the following C program?

```
#include <stdio.h>

int main() {
    double a[2]={20.0, 25.0}, *p, *q;
    p = a;
    q = p + 1;
    printf("%d,%d", (int)(q - p), (int)(*q - *p));
    return 0;}
```

- (A) 4,8
 - (B) 1,5
 - (C) 8,5
 - (D) 1,8
- Organizing Institute: IISc, Bengaluru

Question 55 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.38 (Computer Science & IT)

Q.38

Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

- (A) 10.12.2.0/23
 - (B) 10.12.2.0/24
 - (C) 10.12.0.0/22
 - (D) 10.12.2.0/22
- Organizing Institute: IISc, Bengaluru

Question 56 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.39 (Computer Science & IT)

Q.39

You are given a set V of distinct integers. A binary search tree T is created by inserting all elements of V one by one, starting with an empty tree. The tree T follows the convention that, at each node, all values stored in the left subtree of the node are smaller than the value stored at the node. You are not aware of the sequence in which these values were inserted into T , and you do not have access to T .

Which one of the following statements is TRUE?

- (A) Inorder traversal of T can be determined from V
- (B) Root node of T can be determined from V

- (C) Preorder traversal of T can be determined from V
- (D) Postorder traversal of T can be determined from V

Question 57 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.43 (Computer Science & IT)

Q.43

Consider the following expression: $x[i] = (p + r) - s[i] + u/w$. The following sequence shows the list of triples representing the given expression, with entries missing for triples (1), (3), and (6).

Which one of the following options fills in the missing entries CORRECTLY?

(0)

+

p

r

(1)

(2) uminus (1)

(3)

(4)

/

u

w

(5)

+

(3) (4)

(6)

(7)

=

(6) (5)

- (A) (1) = $(p + r) - s[i]$ (3) * (0) (2) (6) = $(p + r) - s[i] + u/w$
- (B) (1) = $(p + r) - s[i]$ (3) - (0) (2) (6) = $(p + r) - s[i] + u/w$
- (C) (1) = $(p + r) - s[i]$ (3) * (0) (2) (6) = $(p + r) - s[i] + u/w$
- (D) (1) = $(p + r) - s[i]$ (3) - (0) (2) (6) = $(p + r) - s[i] + u/w$

Question 58 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.45 (Computer Science & IT)

Q.45

The relation schema, Person(pid,city), describes the city of residence for every person uniquely identified by pid. The following relational algebra operators are available: selection, projection, cross product, and rename.

To find the list of cities where at least 3 persons reside, using the above operators, the minimum number of cross product operations that must be used is

- (A) 1
- (B) 2
- (C) 3
- (D) 4 Organizing Institute: IISc, Bengaluru

Question 59 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.48 (Computer Science & IT)

Q.48

Let S1 and S2 be two stacks. S1 has capacity of 4 elements. S2 has capacity of 2 elements. S1 already has 4 elements: 100, 200, 300, and 400, whereas S2 is empty, as shown below.

Only the following three operations are available:

PushToS2: Pop the top element from S1 and push it on S2.

PushToS1: Pop the top element from S2 and push it on S1.

GenerateOutput: Pop the top element from S1 and output it to the user.

Note that the pop operation is not allowed on an empty stack and the push operation is not allowed on a full stack.

Which of the following output sequences can be generated by using the above operations?

400 (Top)

300

200

100

Stack S1

Stack S2

- (A) 100, 200, 400, 300
- (B) 200, 300, 400, 100
- (C) 400, 200, 100, 300
- (D) 300, 200, 400, 100 Organizing Institute: IISc, Bengaluru

Question 60 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.49 (Computer Science & IT)

Q.49

Which of the following is/are EQUAL to 224 in radix-5 (i.e., base-5) notation?

- (A) 64 in radix-10
- (B) 100 in radix-8
- (C) 50 in radix-16
- (D) 121 in radix-7 Organizing Institute: IISc, Bengaluru

Question 61 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.50 (Computer Science & IT)

Q.50

Consider 4-variable functions f_1, f_2, f_3, f_4 expressed in sum-of-minterms form as given below.

$$f_1 = \sum(0,2,3,5,7,8,11,13)$$

$$f_2 = \sum(1,3,5,7,11,13, 15)$$

$$f_3 = \sum(0,1,4,11)$$

$$f_4 = \sum(0,2,6,13)$$

With respect to the circuit given above, which of the following options is/are CORRECT?

- (A) $Y = \sum(0,1,2,11,13)$
- (B) $Y = \prod(3,4, 5,6,7,8,9,10,12,14,15)$
- (C) $Y = \sum(0,1,2,3,4,5,6,7)$
- (D) $Y = \prod(8,9,10,11,12,13,14,15)$ Organizing Institute: IISc, Bengaluru

Question 62 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.57 (Computer Science & IT)

Q.57

A processor with 16 general purpose registers uses a 32-bit instruction format. The instruction format consists of an opcode field, an addressing mode field, two register operand fields, and a 16-bit scalar field. If 8 addressing modes are to be supported, the maximum number of unique opcodes possible for every addressing mode is _____

Organizing Institute: IISc, Bengaluru

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 63 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.58 (Computer Science & IT)

Q.58

A non-pipelined instruction execution unit operating at 2 GHz takes an average of 6 cycles to execute an instruction of a program P. The unit is then redesigned to operate on a 5-stage pipeline at 2 GHz. Assume that the ideal throughput of the pipelined unit is 1 instruction per cycle. In the execution of program P, 20% instructions incur an average of 2 cycles stall due to data hazards and 20% instructions incur an average of 3 cycles stall due to control hazards. The speedup (rounded off to one decimal place) obtained by the pipelined design over the non-pipelined design is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 64 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.60 (Computer Science & IT)

Q.60

The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. The chromatic number of the following graph is _____

Organizing Institute: IISc, Bengaluru

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 65 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.61 (Computer Science & IT)

Q.61

A processor uses a 32-bit instruction format and supports byte-addressable memory access. The ISA of the processor has 150 distinct instructions. The instructions are equally divided into two types, namely R-type and I-type, whose formats are shown below.

R-type Instruction Format:

OPCODE UNUSED DST Register

SRC Register1

SRC Register 2

I-type Instruction Format:

OPCODE DST Register SRC Register # Immediate value/address

In the OPCODE, 1 bit is used to distinguish between I-type and R-type instructions and the remaining bits indicate the operation. The processor has 50 architectural registers, and all register fields in the instructions are of equal size.

Let X be the number of bits used to encode the UNUSED field, Y be the number of bits used to encode the OPCODE field, and Z be the number of bits used to encode the immediate value/address field. The value of $X + 2Y + Z$ is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 66 of 122 | GATE 2024 Set 2 (CS2 S6) | Q.63 (Computer Science & IT)

Q.63

Let Z_n be the group of integers $\{0, 1, 2, \dots, n-1\}$ with addition modulo n as the group operation. The number of elements in the group $Z_2 \times Z_3 \times Z_4$ that are their own inverses is _____

Organizing Institute: IISc, Bengaluru

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 67 of 122 | GATE 2023 Single Session | Q.32 (Computer Science & IT)

Q.32

Consider a 3-stage pipelined processor having a delay of 10 ns (nanoseconds), 20 ns, and 14 ns, for the first, second, and the third stages, respectively. Assume that there is no other delay and the processor does not suffer from any pipeline

hazards. Also assume that one instruction is fetched every cycle.

The total execution time for executing 100 instructions on this processor is

ns.

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 68 of 122 | GATE 2023 Single Session | Q.33 (Computer Science & IT)

Q.33

A keyboard connected to a computer is used at a rate of 1 keystroke per second.

The computer system polls the keyboard every 10 ms (milli seconds) to check for a keystroke and consumes 100 μ s (micro seconds) for each poll. If it is determined after polling that a key has been pressed, the system consumes an additional 200 μ s to process the keystroke. Let T_1 denote the fraction of a second spent in polling and processing a keystroke.

In an alternative implementation, the system uses interrupts instead of polling.

An interrupt is raised for every keystroke. It takes a total of 1 ms for servicing an interrupt and processing a keystroke. Let T_2 denote the fraction of a second spent in servicing the interrupt and processing a keystroke.

The ratio T_1

T_2

is

. (Rounded off to one decimal place)

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 69 of 122 | GATE 2023 Single Session | Q.34 (Computer Science & IT)

Q.34

The integer value printed by the ANSI-C program given below is

.

```
#include<stdio.h>
int funcp(){
static int x = 1;
x++;
return x;
}
int main(){
int x,y;
x = funcp();
y = funcp()+x;
printf("%d\n", (x+y));
return 0;
```

```
}  
CS
```

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 70 of 122 | GATE 2023 Single Session | Q.42 (Computer Science & IT)

Q.42

CS

Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning, Q1, Q2 and Q3 have values 0, 1 and 1, respectively.

T

Q

CLK

D

Q

CLK

T

Q

CLK

CLKIN

Q1

Q2

Q3

Which one of the given values of (Q1, Q2, Q3) can NEVER be obtained with this digital circuit?

- (A) (0, 0, 1)
- (B) (1, 0, 0)
- (C) (1, 0, 1)
- (D) (1, 1, 1) CS

Question 71 of 122 | GATE 2023 Single Session | Q.44 (Computer Science & IT)

Q.44

CS

Consider the IEEE-754 single precision floating point numbers
 $P = 0xC1800000$ and $Q = 0x3F5C2EF4$.

Which one of the following corresponds to the product of these numbers
(i.e., $P \times Q$), represented in the IEEE-754 single precision format?

- (A) 0x404C2EF4
- (B) 0x405C2EF4

- (C) 0xC15C2EF4
- (D) 0xC14C2EF4 CS

Question 72 of 122 | GATE 2023 Single Session | Q.63 (Computer Science & IT)

Q.63

An 8-way set associative cache of size 64 KB (1 KB = 1024 bytes) is used in a system with 32-bit address. The address is sub-divided into TAG, INDEX, and BLOCK OFFSET.

The number of bits in the TAG is

.

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 73 of 122 | GATE 2022 Single Session | Q.17 (Computer Science & IT)

Q.17

Which one of the following facilitates transfer of bulk data from hard disk to main memory with the highest throughput?

- (A) DMA based I/O transfer
- (B) Interrupt driven I/O transfer
- (C) Polling based I/O transfer
- (D) Programmed I/O transfer

Question 74 of 122 | GATE 2022 Single Session | Q.18 (Computer Science & IT)

Q.18

Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation $R1 + R2$, which one of the following values of R1 and R2 gives an arithmetic overflow?

- (A) R1 = 1011 and R2 = 1110
- (B) R1 = 1100 and R2 = 1010
- (C) R1 = 0011 and R2 = 0100
- (D) R1 = 1001 and R2 = 1111

Question 75 of 122 | GATE 2022 Single Session | Q.33 (Computer Science & IT)

Q.33

A cache memory that has a hit rate of 0.8 has an access latency 10 ns and miss penalty 100 ns. An optimization is done on the cache to reduce the miss rate.

However, the optimization results in an increase of cache access latency to 15 ns, whereas the miss penalty is not affected. The minimum hit rate (rounded off to two decimal places) needed after the optimization such that it should not increase the average memory access time is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 76 of 122 | GATE 2022 Single Session | Q.35 (Computer Science & IT)

Q.35

Consider the resolution of the domain name www.gate.org.in by a DNS resolver. Assume that no resource records are cached anywhere across the DNS servers and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 77 of 122 | GATE 2022 Single Session | Q.41 (Computer Science & IT)

Q.41

Consider three floating point numbers A, B and C stored in registers RA, RB and RC, respectively as per IEEE-754 single precision floating point format. The 32-bit content stored in these registers (in hexadecimal form) are as follows.

RA = 0xC1400000

RB = 0x42100000

RC = 0x41400000

Which one of the following is FALSE?

- (A) $0 < A < B < C$
- (B) $C < A < B$
- (C) $3 < B < C$
- (D) $(-1) < 0 < B < C$

Question 78 of 122 | GATE 2022 Single Session | Q.45 (Computer Science & IT)

Q.45

Consider solving the following system of simultaneous equations using LU decomposition.

- $$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 14 \\ 12 \\ 20 \end{bmatrix}$$

1
2
3
2
4
3
7
2
5
7
x
x
x
x
x
x
x
x
x
x
x
□
□
□
□
□
□
□
□
□
□
□
where L and U are denoted as
11
21
22
31
32
33
0
0
0
L
L
L
L
L

L
L
□
□
□
,
11
12
13
22
23
33
0
0
0
U
U
U
U
U
U
U
U
□
□
□

Which one of the following is the correct combination of values for

32
L,
33,
U
and
1?
x

- (A) 32 33 1 1 2, , 1 2 L U x □ □ □ □
- (B) 32 33 1 2, 2, 1 L U x □ □ □ □
- (C) 32 33 1 1, 2, 0 2 L U x □ □ □ □
- (D) 32 33 1 1 1, , 0 2 2 L U x □ □ □ □

Question 79 of 122 | GATE 2022 Single Session | Q.54 (Computer Science & IT)

Q.54

Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes.

The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (i.e., PQRSPQRS...). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

- (A) Every access to S is a hit.
- (B) Once P is brought to the cache it is never evicted.
- (C) At the end of the execution only R and S reside in the cache.
- (D) Every access to R evicts Q from the cache.

Question 80 of 122 | GATE 2022 Single Session | Q.61 (Computer Science & IT)

Q.61

A processor X1 operating at 2 GHz has a standard 5-stage RISC instruction pipeline having a base CPI (cycles per instruction) of one without any pipeline hazards. For a given program P that has 30% branch instructions, control hazards incur 2 cycles stall for every branch. A new version of the processor X2 operating at same clock frequency has an additional branch predictor unit (BPU) that completely eliminates stalls for correctly predicted branches. There is neither any savings nor any additional stalls for wrong predictions. There are no structural hazards and data hazards for X1 and X2. If the BPU has a prediction accuracy of 80%, the speed up (rounded off to two decimal places) obtained by X2 over X1 in executing P is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 81 of 122 | GATE 2022 Single Session | Q.63 (Computer Science & IT)

Q.63

Consider two files systems A and B, that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are

A
n and
,
B

n respectively, then the value of

A

B

n

n

□

is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 82 of 122 | GATE 2021 Set 1 (CS1) | Q.1 (Computer Science & IT)

Q.1

The ratio of boys to girls in a class is 7 to 3.

Among the options below, an acceptable value for the total number of students in the class is:

(A) 21

(B) 37

(C) 50

(D) 73

Question 83 of 122 | GATE 2021 Set 2 (CS2) | Q.4 (Computer Science & IT)

Q.4

If

2

2

1

3

2

2

2

x

x

x

□

□

–

–

–

=

+

□

□

, then the value of x is:

- (A) 2
- (B) 4
- (C) 6
- (D) 8

Question 84 of 122 | GATE 2019 Single Session | Q.1 (Computer Science & IT)

Q.1

A certain processor uses a fully associative cache of size 16 kB. The cache block size is 16 bytes. Assume that the main memory is byte addressable and uses a 32-bit address. How many bits are required for the Tag and the Index fields respectively in the addresses generated by the processor?

- (A) 24 bits and 0 bits
- (B) 28 bits and 4 bits
- (C) 24 bits and 4 bits
- (D) 28 bits and 0 bits

Question 85 of 122 | GATE 2019 Single Session | Q.14 (Computer Science & IT)

Q.14 Which one of the following statements is NOT correct about the B+ tree data structure used for creating an index of a relational database table?

- (A) B+ Tree is a height-balanced tree
- (B) Non-leaf nodes have pointers to data records
- (C) Key values in each node are kept in sorted order
- (D) Each leaf node has a pointer to the next leaf node

Question 86 of 122 | GATE 2019 Single Session | Q.18 (Computer Science & IT)

Q.18 Consider the following C program:

```
#include <stdio.h>
int jumble(int x, int y){
    x=2*x+y;
    return x;
}
int main(){
    int x=2, y=5;
    y=jumble(y,x);
```

```

x=jumble(y,x);
printf("%d \n", x);
return 0;
}

```

The value printed by the program is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 87 of 122 | GATE 2019 Single Session | Q.21 (Computer Science & IT)

Q.21 The value of $351 \bmod 5$ is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 88 of 122 | GATE 2019 Single Session | Q.23 (Computer Science & IT)

Q.23 Consider three concurrent processes P1, P2 and P3 as shown below, which access a shared variable D that has been initialized to 100.

```

P1
P2
P3
:
:
D = D + 20
:
:
:
:
D = D - 50
:
:
:
:
D = D + 10
:
:

```

The processes are executed on a uniprocessor system running a time-shared operating system. If the minimum and maximum possible values of D after the three processes have completed execution are X and Y respectively, then the value of $Y - X$ is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 89 of 122 | GATE 2019 Single Session | Q.27 (Computer Science & IT)

Q.27 Consider the following C program:

```
#include <stdio.h>
```

```

int r(){
static int num=7;
return num--;
}
int main(){
for (r();r();r())
printf("%d",r());
return 0;
}

```

Which one of the following values will be displayed on execution of the programs?

- (A) 41
- (B) 52
- (C) 63
- (D) 630 CS 7/15

Question 90 of 122 | GATE 2019 Single Session | Q.45 (Computer Science & IT)

Q.45 A certain processor deploys a single-level cache. The cache block size is 8 words and the word size is 4 bytes. The memory system uses a 60-MHz clock. To service a cache miss, the memory controller first takes 1 cycle to accept the starting address of the block, it then takes 3 cycles to fetch all the eight words of the block, and finally transmits the words of the requested block at the rate of 1 word per cycle. The maximum bandwidth for the memory system when the program running on the processor issues a series of read operations is _____ $\times$ 106 bytes/sec.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 91 of 122 | GATE 2019 Single Session | Q.46 (Computer Science & IT)

Q.46 Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.) Suppose two leaves a and b of T are chosen uniformly and independently at random. The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 92 of 122 | GATE 2019 Single Session | Q.54 (Computer Science & IT)

Q.54 In an RSA cryptosystem, the value of the public modulus parameter n is 3007. If it is also known that $\phi(n) = 2880$, where $\phi()$ denotes Euler's Totient Function, then the prime factor of n which is greater than 50 is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 93 of 122 | GATE 2018 Single Session | Q.5 (Computer Science & IT)

Q.5

Consider the following processor design characteristics.

I.

Register-to-register arithmetic operations only

II.

Fixed-length instruction format

III.

Hardwired control unit

Which of the characteristics above are used in the design of a RISC processor?

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) I, II and III

Question 94 of 122 | GATE 2018 Single Session | Q.9 (Computer Science & IT)

Q.9

The following are some events that occur after a device controller issues an interrupt while process L is under execution.

(P) The processor pushes the process status of L onto the control stack.

(Q) The processor finishes the execution of the current instruction.

(R) The processor executes the interrupt service routine.

(S) The processor pops the process status of L from the control stack.

(T) The processor loads the new PC value based on the interrupt.

Which one of the following is the correct order in which the events above occur?

- (A) QPTRS
- (B) PTRSQ
- (C) TRPQS
- (D) QTPRS

Question 95 of 122 | GATE 2018 Single Session | Q.22 (Computer Science & IT)

Q.22 Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops.

clock

The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of "in" is ____.

in

D Q

out
D Q
CS
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[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 96 of 122 | GATE 2018 Single Session | Q.32 (Computer Science & IT)

Q.32 Consider the following C code. Assume that unsigned long int type length is 64 bits.

```
unsigned long int fun(unsigned long int n){
    unsigned long int i, j = 0, sum = 0;
    for (i = n; i > 1; i = i/2) j++;
    for (; j > 1; j = j/2) sum++;
    return(sum);
}
```

The value returned when we call fun with the input 240 is

- (A) 4
- (B) 5
- (C) 6
- (D) 40

Question 97 of 122 | GATE 2018 Single Session | Q.34 (Computer Science & IT)

Q.34 The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N bytes. The size of each cache block is 2^M words. For a K -way set-associative cache memory, the length (in number of bits) of the tag field is

- (A) $P - N - \log_2 K$
- (B) $P - N + \log_2 K$
- (C) $P - N - M - W - \log_2 K$
- (D) $P - N - M - W + \log_2 K$

Question 98 of 122 | GATE 2018 Single Session | Q.41 (Computer Science & IT)

Q.41 Consider the relations $r(A, B)$ and $s(B, C)$, where $s.B$ is a primary key and $r.B$ is a foreign key referencing $s.B$. Consider the query

Q: $r \bowtie (\sigma_{B < 5}(s))$

Let LOJ denote the natural left outer-join operation. Assume that r and s contain no null values.

Which one of the following queries is NOT equivalent to Q?

- (A) $\sigma_{B < 5}(r \bowtie s)$
- (B) $\sigma_{B < 5}(r \text{ LOJ } s)$
- (C) $r \text{ LOJ } (\sigma_{B < 5}(s))$
- (D) $\sigma_{B < 5}(r \text{ LOJ } s \text{ CS } 15/20)$

Question 99 of 122 | GATE 2018 Single Session | Q.42 (Computer Science & IT)

Q.42 Consider the following four relational schemas. For each schema, all non-trivial functional dependencies are listed. The underlined attributes are the respective primary keys.

Schema I: Registration (rollno, courses)

Field 'courses' is a set-valued attribute containing the set of courses a student has registered for.

Non-trivial functional dependency:

rollno $\rightarrow$ courses

Schema II: Registration (rollno, courseid, email)

Non-trivial functional dependencies:

rollno, courseid $\rightarrow$ email

email $\rightarrow$ rollno

Schema III: Registration (rollno, courseid, marks, grade)

Non-trivial functional dependencies:

rollno, courseid $\rightarrow$ marks, grade

marks $\rightarrow$ grade

Schema IV: Registration (rollno, courseid, credit)

Non-trivial functional dependencies:

rollno, courseid $\rightarrow$ credit

courseid $\rightarrow$ credit

Which one of the relational schemas above is in 3NF but not in BCNF?

- (A) Schema I
- (B) Schema II
- (C) Schema III
- (D) Schema IV

Question 100 of 122 | GATE 2018 Single Session | Q.46 (Computer Science & IT)

Q.46 The number of possible min-heaps containing each value from {1, 2, 3, 4, 5, 6, 7} exactly once is ____.

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[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 101 of 122 | GATE 2018 Single Session | Q.47 (Computer Science & IT)

Q.47 Consider the following undirected graph G:

Choose a value for x that will maximize the number of minimum weight spanning trees (MWSTs) of G. The number of MWSTs of G for this value of x is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 102 of 122 | GATE 2018 Single Session | Q.48 (Computer Science & IT)

Q.48 Consider the weights and values of items listed below. Note that there is only one unit of each item.

Item number

Weight

(in Kgs)

Value

(in Rupees)

1	10	60
2	7	28
3	4	20
4	2	24

The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt} . A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy} .

The value of $V_{opt} - V_{greedy}$ is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 103 of 122 | GATE 2018 Single Session | Q.50 (Computer Science & IT)

Q.50 The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB).

The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the PO stage, 40 instructions take 3 clock cycles each, 35

instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instructions is ____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 104 of 122 | GATE 2018 Single Session | Q.51 (Computer Science & IT)

Q.51 A processor has 16 integer registers ($R_0, R_1, \dots, R_{15}$) and 64 floating point registers ($F_0, F_1, \dots, F_{63}$). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type-4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating point register operand (1F).

The maximum value of N is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 105 of 122 | GATE 2018 Single Session | Q.54 (Computer Science & IT)

Q.54 Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and a 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0.

The fragmentation offset value stored in the third fragment is ____.

CS

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[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 106 of 122 | GATE 2016 Set 1 (CS1) | Q.14 (Computer Science & IT)

Q.14

Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are TRUE?

P: Minimum spanning tree of G does not change

Q: Shortest path between any pair of vertices does not change

- (A) P only
- (B) Q only
- (C) Neither P nor Q

(D) Both P and Q CS(Set A) 4/17 GATE 2016

Question 107 of 122 | GATE 2016 Set 1 (CS1) | Q.27 (Computer Science & IT)

Q.27

Consider the recurrence relation $a_1 = 8$, $a_n = 6n^2 + 2n + a_{n-1}$. Let $a_{99} = K \times 104$. The value of K is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 108 of 122 | GATE 2016 Set 1 (CS1) | Q.28 (Computer Science & IT)

Q.28

A function $f: \mathbb{N}^+ \rightarrow \mathbb{N}^+$, defined on the set of positive integers $\mathbb{N}^+$, satisfies the following properties:

$$f(n) = f(n/2)$$

if n is even

$$f(n) = f(n+5)$$

if n is odd

Let $R = \{i | \exists j : f(j) = i\}$ be the set of distinct values that f takes. The maximum possible size of R is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 109 of 122 | GATE 2016 Set 1 (CS1) | Q.31 (Computer Science & IT)

Q.31

The size of the data count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29,154 kilobytes from disk to main memory. The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 110 of 122 | GATE 2016 Set 1 (CS1) | Q.32 (Computer Science & IT)

Q.32

The stage delays in a 4-stage pipeline are 800, 500, 400 and 300 picoseconds. The first stage (with delay 800 picoseconds) is replaced with a functionally equivalent design involving two stages with respective delays 600 and 350 picoseconds. The throughput increase of the pipeline is percent.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 111 of 122 | GATE 2016 Set 1 (CS1) | Q.34 (Computer Science & IT)

Q.34

The following function computes the maximum value contained in an integer array $p[]$ of size n ($n \geq 1$).

```
int max(int *p, int n) {
    int a=0, b=n-1;
    while (_____) {
        if (p[a] <= p[b]) { a = a+1; }
        else
            { b = b-1; }
    }
    return p[a];
}
```

The missing loop condition is

- (A) $a \neq n$
- (B) $b \neq 0$
- (C) $b > (a + 1)$
- (D) $b \neq a$

Question 112 of 122 | GATE 2016 Set 1 (CS1) | Q.45 (Computer Science & IT)

Q.45

The attributes of three arithmetic operators in some programming language are given below.

Operator

Precedence

Associativity

Arity

+

High

Left

Binary

-

Medium

Right

Binary

□

Low

Left

Binary

The value of the expression $2-5+1-7\Box 3$ in this language is

CS(Set A)

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[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 113 of 122 | GATE 2016 Set 2 (CS2) | Q.7 (Computer Science & IT)

Q.7

Consider an eight-bit ripple-carry adder for computing the sum of A and B, where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 114 of 122 | GATE 2016 Set 2 (CS2) | Q.11 (Computer Science & IT)

Q.11

Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n-th vertex in this BFS traversal, then the maximum possible value of n is

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 115 of 122 | GATE 2016 Set 2 (CS2) | Q.12 (Computer Science & IT)

Q.12

The value printed by the following program is

```
.  
void f(int* p, int m){  
    m = m + 5;  
    *p = *p + m;  
    return;  
}  
void main(){  
    int i=5, j=10;  
    f(&i, j);  
    printf("%d", i+j);  
}
```

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 116 of 122 | GATE 2016 Set 2 (CS2) | Q.29 (Computer Science & IT)

Q.29

The value of the expression $1399 \pmod{17}$, in the range 0 to 16, is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 117 of 122 | GATE 2016 Set 2 (CS2) | Q.31 (Computer Science & IT)

Q.31

Consider a processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and a twelve-bit immediate value. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 100 instructions, the amount of memory (in bytes) consumed by the program text is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 118 of 122 | GATE 2016 Set 2 (CS2) | Q.32 (Computer Science & IT)

Q.32

The width of the physical address on a machine is 40 bits. The width of the tag field in a 512 KB 8-way set associative cache is

bits.

CS(Set B)

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[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 119 of 122 | GATE 2016 Set 2 (CS2) | Q.33 (Computer Science & IT)

Q.33

Consider a 3 GHz (gigahertz) processor with a three-stage pipeline and stage latencies τ_1 , τ_2 , and τ_3 such that $\tau_1 = 3\tau_2/4 = 2\tau_3$. If the longest pipeline stage is split into two pipeline stages of equal latency, the new frequency is

GHz, ignoring delays in the pipeline registers.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 120 of 122 | GATE 2016 Set 2 (CS2) | Q.37 (Computer Science & IT)

Q.37

Consider the following program:

```
int f(int *p, int n)
{
```

```

if (n <= 1) return 0;
else return max(f(p+1,n-1),p[0]-p[1]);
}
int main()
{
int a[] = {3,5,2,6,4};
printf("%d", f(a,5));
}

```

Note: $\max(x,y)$ returns the maximum of x and y .

The value printed by this program is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 121 of 122 | GATE 2016 Set 2 (CS2) | Q.39 (Computer Science & IT)

Q.39

The given diagram shows the flowchart for a recursive function $A(n)$.

Assume that all

statements, except for the recursive calls, have $O(1)$ time complexity. If the worst case time complexity of this function is $O(n^\alpha)$, then the least possible value (accurate up to two decimal positions) of α is

.

Start

$A(n/2)$

$A(n/2)$

Return

$A(n/2)$

$A(n/2)$

Return

$A(n/2)$

Return

$A(n/2)$

Return

Flowchart for Recursive Function $A(n)$

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 122 of 122 | GATE 2016 Set 2 (CS2) | Q.50 (Computer Science & IT)

Q.50

A file system uses an in-memory cache to cache disk blocks. The miss rate of the cache is shown in the figure. The latency to read a block from the cache is 1 ms and to read a block from the disk is 10 ms. Assume that the cost of checking whether a block exists in the cache is negligible. Available cache sizes are in multiples of 10 MB.

10 20 30 40 50

10

60 70 80 90

20

30

40

50

60

70

80

90

0

0

Miss rate (%)

Cache size (MB)

The smallest cache size required to ensure an average read latency of less than 6 ms is MB.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]